THYROID AND PARATHYROID GLANDS

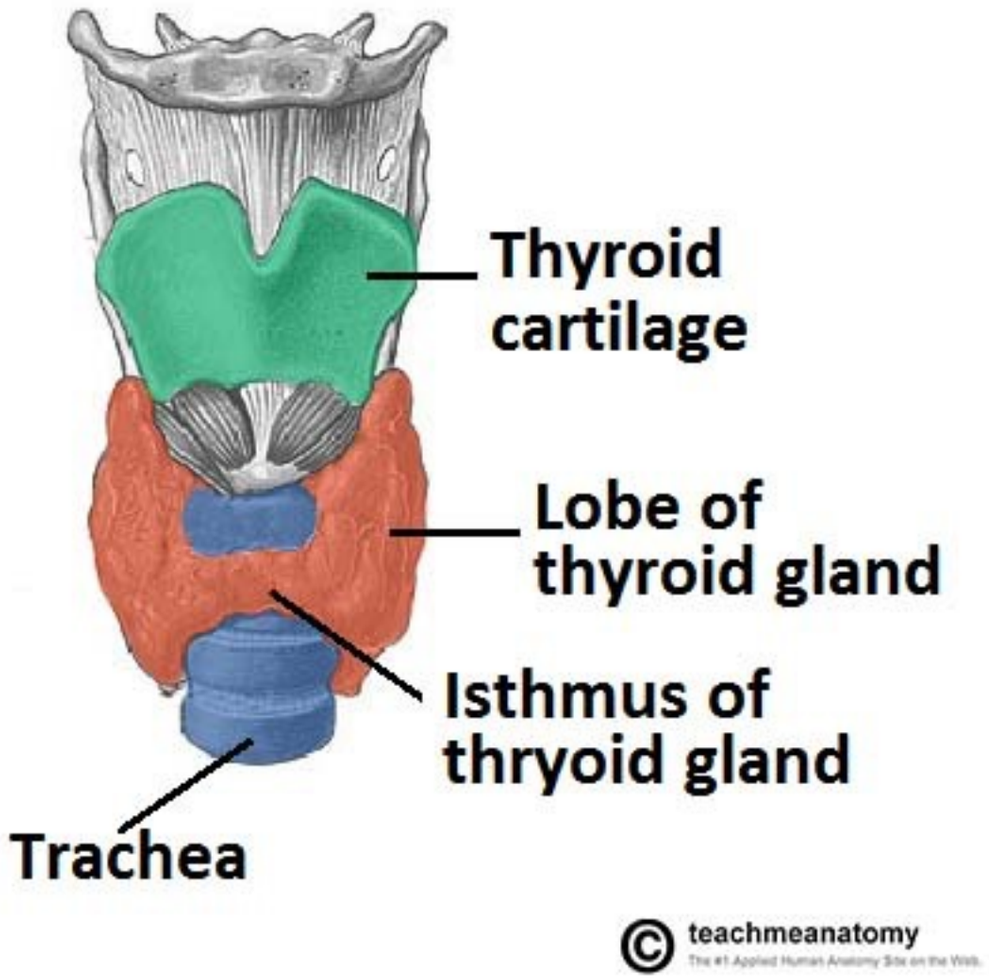
LEARNING OBJECTIVES

- to understand gross anatomy of thyroid and parathyroid glands

THYROID GLAND

- shield like having rich blood supply, situated in lower part and front of neck
- regulates BMR, stimulate somatic and psychic growth and play important role in calcium metabolism
- only gland using natural iodine for synthesis of hormones
- right and left lobes connected by isthmus

LOCATION



- The thyroid gland is located in the **anterior neck**, spanning between the C5 and T1 vertebrae.
- It lies behind the sternohyoid and sternothyroid muscles, **wrapping** around the cricoid cartilage and superior tracheal rings. It is inferior to the thyroid cartilage of the **larynx**. The gland is in the **visceral compartment** of the neck, along with the trachea, oesophagus and pharynx. The compartment is bound by pretracheal fascia.
- each lobe extends from middle of thyroid cartilage to fourth or fifth of tracheal rings
- isthmus extends extends from second to fourth tracheal rings

CAPSULES

- a true capsule is peripheral condensation of connective tissue of gland
- dense capillary plexus is present deep to true capsule
- false capsule is derived from pretracheal layer of deep cervical fascia
- it is thin along posterior border of lobes but thick on inner surface of gland forming suspensory ligament of Berry, connecting the lobe to cricoid cartilage

PARTS AND RELATIONS

- lobes are conical having
  - apex,
  - base,
  - three surfaces (lateral/medial/posterolateral)
  - two borders (anterior/ posterior)

- apex is directed upwards and slightly lateral, limited superiorly by attachment of sternothyroid muscle to oblique line of thyroid cartilage which is medial to apex. Apex is related to superior thyroid artery and external laryngeal nerve
- base is at level with 4th or 5th tracheal ring related to inferior thyroid artery and recurrent laryngeal nerve
- lateral or superficial surface is convex and covered by:
  - sternohyoid
  - superior belly of omohyoid
  - sternothyroid
  - anterior border of sternocleidomastoid

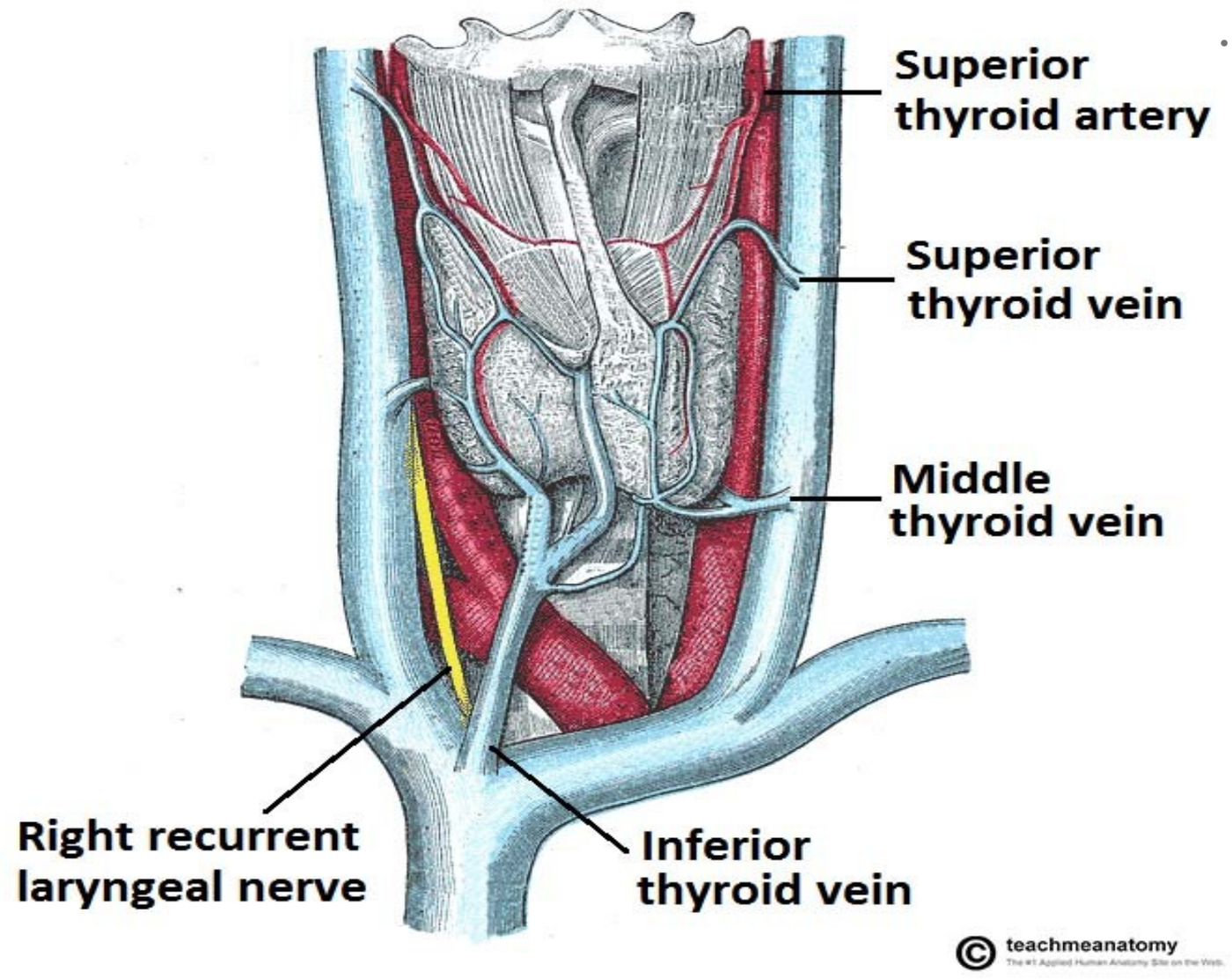
- the medial surface is related to:
  - two tubes (trachea and oesophagus)
  - two muscles (inferior constrictor and cricothyroid)
  - two nerves (external laryngeal and recurrent laryngeal)

  - Posterior surface is related to carotid sheath and overlaps common carotid artery
  - anterior border is thin and related to anterior branch of superior thyroid artery
  - posterior border is thick and rounded and separates the medial and posterior surfaces

- posterior border relates to
  - inferior thyroid artery
  - anastomosis between posterior branch of superior and ascending branch of inferior thyroid arteries
  - parathyroid glands
  - thoracic duct only on left side

- isthmus connects lower parts of two lobes. it has:
  - two surfaces: Anterior and Posterior
  - Two borders: Superior and Inferior
  - Anterior surface is covered by
    - right and left sternothyroid and sternohyoid muscles
    - the anterior jugular veins
    - fascia and skin
  - Posterior surface is related to 2nd to 4th tracheal rings
  - upper border relates to anterior branches of right and left superior thyroid arteries which anastomose here
  - at lower border inferior thyroid veins leave the gland

VASCULAR SUPPLY



- The thyroid gland secretes hormones directly into the blood. Therefore it needs to be **highly vascularised**. Blood supply to the thyroid gland is achieved by two main arteries; the superior and inferior thyroid arteries. These are **paired** arteries arising on both the left and right.
- The **superior thyroid artery** is the first branch of the external carotid artery. After arising, the artery descends toward the thyroid gland. As a generalisation, it supplies the **superior and anterior** portions of the gland.
- The **inferior thyroid artery** arises from the thyrocervical trunk (which in turn is a branch of the subclavian artery). The artery travels superomedially to reach the inferior pole of the thyroid. It tends to supply the **postero-inferior** aspect.
- In a small proportion of people (around 10%), there is an additional artery present; the **thyroid ima artery**. It comes from the brachiocephalic trunk of the arch of aorta, supplying the anterior surface and isthmus.
- Venous drainage is carried out by the superior, middle and inferior thyroid veins, which form a **venous plexus**. The superior and middle veins drain into the **internal jugular veins**, whereas the inferior drains into the **brachiocephalic** vein.

LYMPHATICS

- lymph from upper part of gland reaches deep cervical LN either directly or through prelaryngeal LN
- lymph from lower part of gland drains to lower deep cervical nodes directly and also through pretracheal and paratracheal LN

NERVE SUPPLY

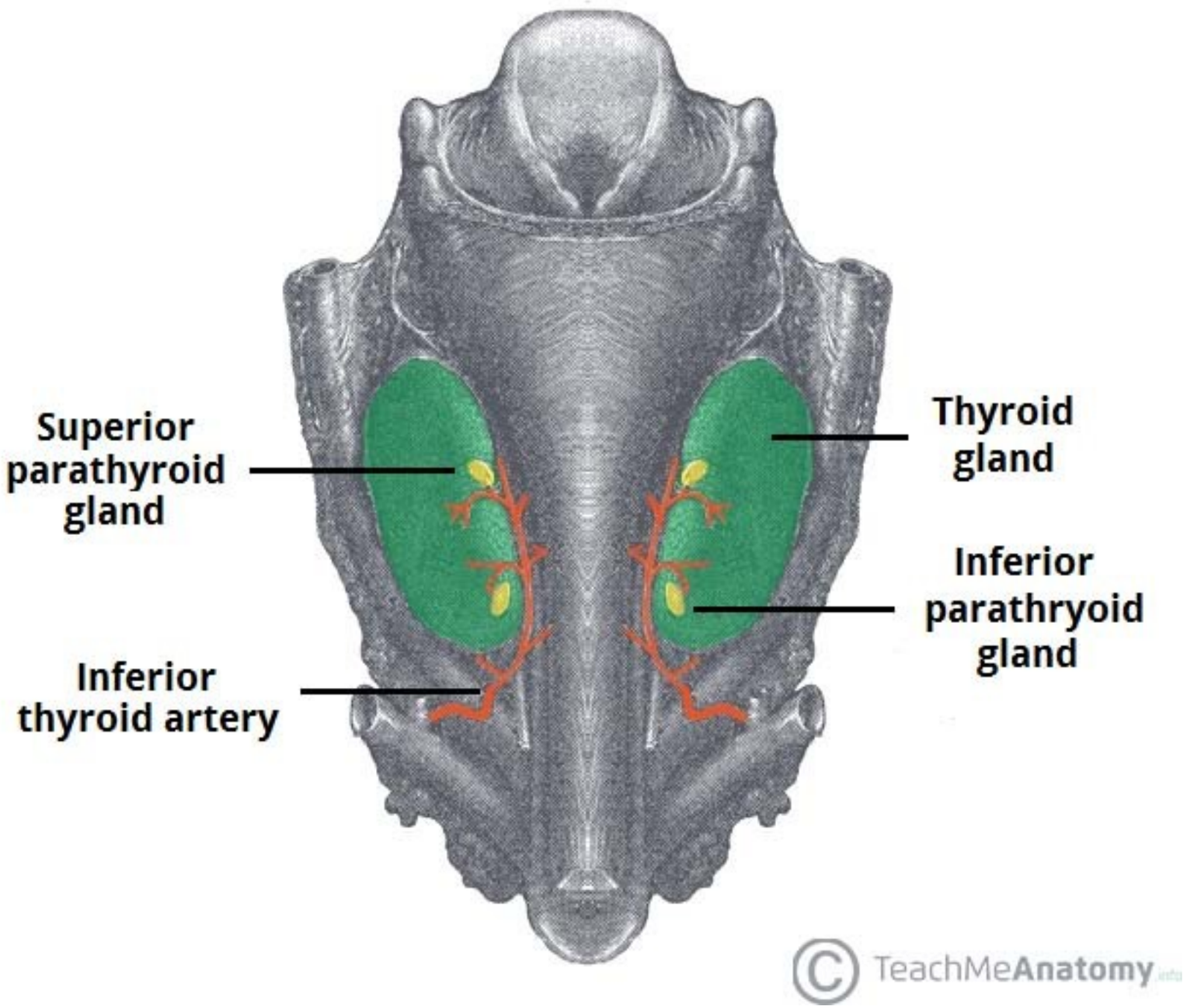
- derived mainly from middle cervical ganglion and partly from superior and inferior cervical ganglia
- they are vasoconstrictor

PARATHYROID GLANDS

- The parathyroid glands are small endocrine glands located in the anterior neck. They are responsible for the production of **parathyroid hormone**, which acts to control calcium levels in the body.
- two pairs, lie on posterior border of thyroid gland
- secrete parathormone control metabolism of calcium and phosphorus along with thyrocalcitonin

POSITION

- anastomosis between superior and inferior thyroid arteries is a good guide to gland because they lie close to it
- superior parathyroid is more constant and lies at middle of posterior border of lobe of thyroid gland. dorsal to recurrent laryngeal nerve
- inferior is variable in position. it may lie:
  - within thyroid capsule below inferior thyroid artery and near lower pole of thyroid lobe
  - behind and outside the thyroid capsule immediately above inferior thyroid artery
  - within substance of lobe near its posterior border. usually ventral to recurrent laryngeal nerve


Superior parathyroid gland
Thyroid gland
Inferior parathryoid gland
Inferior thyroid artery
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VASCULATURE

- The posterior aspect of the thyroid gland is supplied by the **inferior thyroid arteries**. Thus its branches also supply the nearby parathyroid glands. Collateral circulation is delivered by the superior thyroid arteries, thyroid ima artery, and laryngeal, tracheal and oesophageal arteries.
- The parathyroid veins drain into the **thyroid plexus of veins**.

LYMPHATICS

- The lymphatic vessels of the parathyroid glands drain (along with those of the thyroid gland) into the **deep cervical** lymph nodes and **paratracheal** lymph nodes.

NERVE SUPPLY

- The parathyroid glands have an extensive supply of nerves, derived from **thyroid branches** of the cervical (sympathetic) ganglia.
- It is important to note that these nerves are **vasomotor**, not secretomotor – endocrine secretion of parathyroid hormone is controlled hormonally.

THANK YOU